ASSIGNMENT-1

REPORT

1. INTRODUCTION

In this assignment, a classification pipeline was implemented using the MNIST handwritten digit dataset. Only digits 0, 1, and 2 were considered. The goal was to estimate class-wise Gaussian parameters using Maximum Likelihood Estimation (MLE) and classify samples using Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA). Additionally, t-SNE visualization was used to analyse the feature distribution of the dataset.

2. DATASET

The MNIST dataset was downloaded from Kaggle in IDX binary format (as given in the assignment.). Each image was converted into a **784-dimensional feature vector** by flattening the pixel matrix. For this assignment, only digits **0, 1, and 2** were selected.

From each class:

- **100 samples were randomly selected from the training set**
- **100 samples were randomly selected from the test set**

This resulted in **300 training samples and 300 test samples**.

3. MAXIMUM LIKELIHOOD ESTIMATION

For each class $c \in \{0,1,2\}$, the mean vector and covariance matrix were estimated using MLE.

Mean Vector

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} x_i$$

Covariance Matrix

$$\Sigma_c = \frac{1}{N_c} \sum_{i=1}^{N_c} (x_i - \mu_c)(x_i - \mu_c)^T$$

The class-conditional probability distribution was assumed to be a multivariate Gaussian:

$$P(\mathbf{x}|y=c) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}_c|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1}(\mathbf{x}-\boldsymbol{\mu}_c)\right).$$

4. LINEAR DISCRIMINANT ANALYSIS (LDA)

The LDA discriminant function is:

$$g_c(x) = x^T\Sigma^{-1}\mu_c - \frac{1}{2}\mu_c^T\Sigma^{-1}\mu_c + \log P(c)$$

5. QUADRATIC DISCRIMINANT ANALYSIS (QDA)

The QDA discriminant function is:

$$g_c(x) = -\frac{1}{2}\log|\Sigma_c| - \frac{1}{2}(x-\mu_c)^T\Sigma_c^{-1}(x-\mu_c) + \log P(c)$$

6. ACCURACY METRIC

Accuracy is a commonly used metric in classification tasks, measuring the proportion of correct predictions over the total number of predictions.

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\hat{y}_i = y_i) \tag{2}$$

where $\mathbb{1}(\hat{y}_i = y_i)$ is an indicator function that returns 1 if the prediction is correct and 0 otherwise.

7. RESULTS

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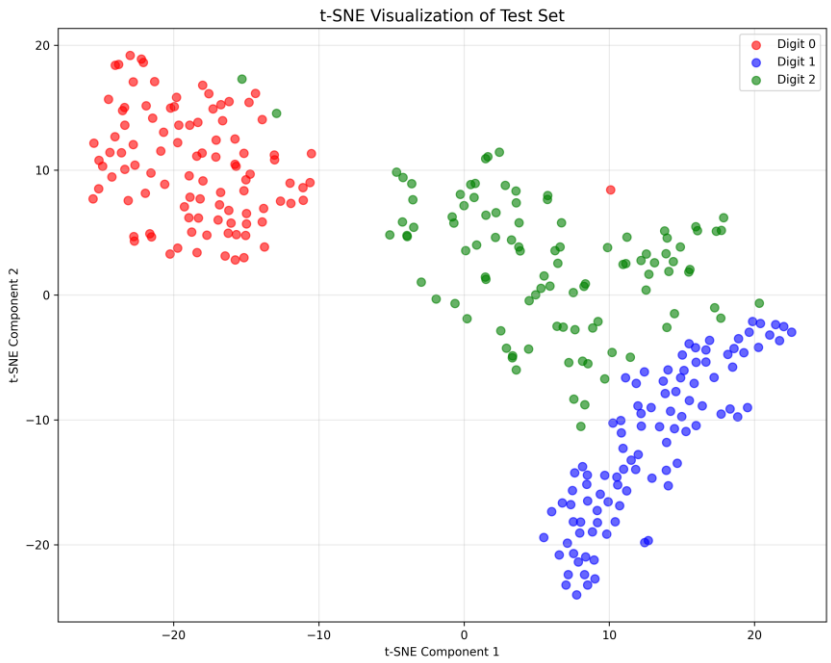
=== LDA Results ===
Overall LDA Accuracy = 0.9033333333333333
Class 0 Accuracy = 0.9600
Class 1 Accuracy = 0.9700
Class 2 Accuracy = 0.7800

=== QDA Results ===
Overall QDA Accuracy = 0.9933333333333333
Class 0 Accuracy = 1.0000
Class 1 Accuracy = 1.0000
Class 2 Accuracy = 0.9800
```

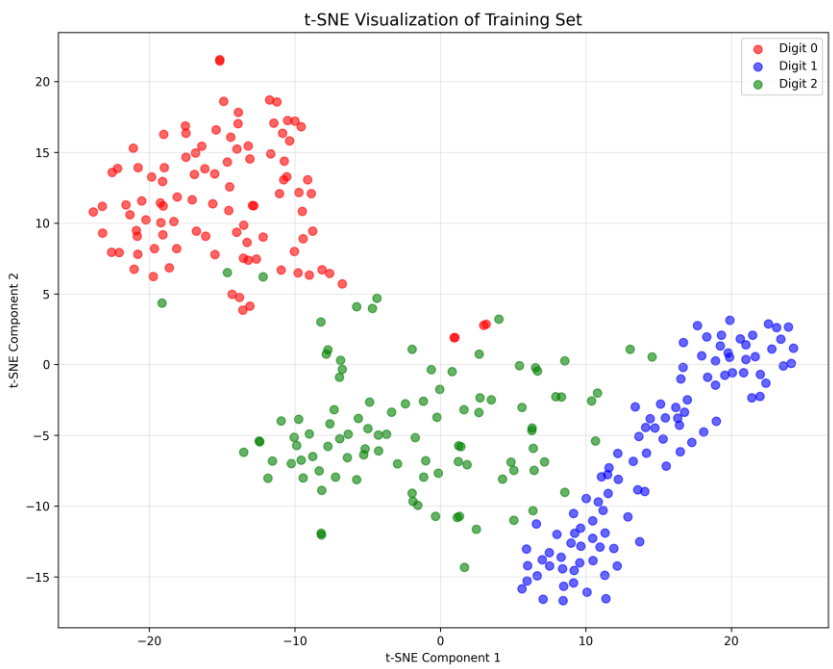
The LDA classifier achieved an overall accuracy of 90.33%, while QDA achieved 99.33%. QDA performed better because it models class-specific covariance matrices, capturing complex class distributions. LDA assumes equal covariance across classes, which led to reduced performance, especially for digit 2. The results indicate that the MNIST digits 0, 1, and 2 are highly separable using Gaussian generative models.

8. t-SNE Visualisation

(i) t-SNE Test Plot



(ii) t-SNE Train Plot



The plots showed that digits 0, 1, and 2 form distinct clusters, indicating that the classes are separable in feature space.

9. DISCRIMINANT VALUES

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=== Discriminant Values for Sample Test Points ===

Test Point Index: 0, True Label: 0
LDA Discriminant Values:
  Class 0: 296.0061
  Class 1: -379.5722
  Class 2: -146.3280
  Predicted Class (LDA): 0
QDA Discriminant Values:
  Class 0: -3706.7867
  Class 1: -42529.6648
  Class 2: -11690.3043
  Predicted Class (QDA): 0

Test Point Index: 50, True Label: 0
LDA Discriminant Values:
  Class 0: 140.5356
  Class 1: -218.7968
  Class 2: 29.1727
  Predicted Class (LDA): 0
QDA Discriminant Values:
  Class 0: -2200.1150
  Class 1: -35955.2829
  Class 2: -9103.6656
  Predicted Class (QDA): 0

Test Point Index: 150, True Label: 1
LDA Discriminant Values:
  Class 0: -219.3558
  Class 1: 318.4122
  Class 2: 86.4507
  Predicted Class (LDA): 1
QDA Discriminant Values:
  Class 0: -10698.9072
  Class 1: -706.0363
  Class 2: -7815.2930
  Predicted Class (QDA): 1
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